TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders

Words: at least ≥ · at most ≤ · more than > · fewer than <.

Solve like an equation: undo + and −, then × and ÷.

Flip the sign when you multiply or divide both sides by a negative number.

Graph: ○ for < and >, ● for ≤ and ≥. Shade right for > and ≥, left for < and ≤.

1. Maya has $40 and earns $12 per hour. Which equation gives the total, y, she has after working x hours?

> **Tip:** Rate goes with x. The start is b.

A. $y = 40x + 12$

B. $y = -12x + 40$

C. $\mathbf{y = 12x + 40}$

Start $40, rate $12 per hour

2. Jordan buys 4 notebooks that cost $1.75 each and some pens for $0.90 each. Write an equation for the total cost, y, when Jordan buys x pens.

> **Tip:** Find the fixed cost first: 4 notebooks × $1.75.

Fixed cost: 4 × 1.75 = _____

y = _____x + _____

$y = 0.90x + 7$

Notebooks: 4 · 1.75 = 7; pens: $0.90x$

3. $f(b) = 3.75b + 25$. Write an inequality for sales of at most $1150, solve it, and graph the solution on a number line.

> **Tip:** At most → ≤. ≤ gets a closed circle.

3.75b + 25 _____ 1150

b _____ _____

Circle: open / closed. Shade: left / right

$b \leq 300$, closed circle at 300, shaded left

$3.75b + 25 \leq 1150 \rightarrow 3.75b \leq 1125 \rightarrow b \leq 300$